

Traffic Incident Prediction & Road Safety Analysis

STAT 4604: Statistical Methods for Engineers

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Traffic Incident Prediction & Road Safety Analysis in Montgomery County, Virginia

Study Objectives: To examine environmental, temporal, road-level, and behavioural predictors of crash severity in Montgomery County, Virginia.

Data: 20,162 crash records from the Virginia DMV /VDOT (data.virginia.gov), covering Montgomery County, VA. The dataset ranges from 2017 – 2025.

Methods: Four-phase analytical approach: exploratory data analysis and visualization, one-way ANOVA and multiple linear regression, ordinal logistic regression (proportional odds model), and supplementary spatial analysis conducted using kernel density estimation.

Key Results: Alcohol-involved crashes had nearly three times the odds of higher severity ($OR = 2.91, p < 0.001$). Surprisingly, night-time driving was associated with reduced severity ($OR = 0.807, p < 0.001$). Spatial analysis revealed that crashes are strongly clustered and not randomly distributed.

Conclusions: Crash severity is driven by a combination of behavioral and environmental conditions. Crash locations are strongly clustered around the US-460/I-81 traffic corridor, the VT campus perimeter and Radford city limits.

INTRODUCTION

Road safety and crash severity is a major challenge facing public health policy makers due to its prevalence, costs, and major personal impact to the people involved. Predicting the severity of crashes based on different factors thus provides us with a unique insight into what leads to crashes, and the steps we can take to prevent them. I chose a localised dataset of Montgomery County to present an exclusive analytical context: a mid-sized Virginia county anchored by a major research university (Virginia Tech), with high traffic volume corridors serving both commuter and student populations.

Prior literature¹ has established that environmental conditions (weather, lighting, road surface) and driver behaviour (alcohol, speeding, distraction) are associated with crash severity. However, most national-scale studies do not capture institution-specific temporal patterns – semester calendars, move-in weekends, game-day traffic – that modulate local crash risk. The primary research question that I will explore is as follows: What environmental, temporal, and road-level factors best predict the severity of traffic incidents in Montgomery County, Virginia? As supplementary analysis, I will also study how proximity to the Virginia Tech campus modulates crash risk. The answer to this question will be motivated by 2 sub-questions: What environmental, temporal, and behavioural conditions are

most predictive of crash severity? Are crashes spatially clustered around high-traffic corridors, and is that clustering statistically non-random?

DATA

The dataset for this study was obtained from the Virginia DMV/VDOT via the Virginia Open Data Portal (<https://data.virginia.gov/dataset/crash-data>). The final dataset, post-processing, contains 20,162 crash records localised to Montgomery County, VA, each with GPS coordinates, crash date and time, severity classification, weather, lighting, road surface conditions, collision type, and binary driver behaviour flags (under the influence of alcohol, speeding, distractions involved).

Severity of crashes was encoded as an ordinal outcome: PDO (property damage only), Injury (to person(s) involved in the crash), and Fatality. For purposes of regression and one-way ANOVA, a numeric severity score was assigned to each severity class: PDO = 1, Injury = 2, Fatal = 3. Binary predictors for driver behavior were also derived from the dataset: *is_dark* (lighting present on road), *is_adverse_weather* (rain/snow/fog/mist), *is_adverse_surface* (road conditions: wet/snowy/slushy), *is_night* (for crashes occurring between 10pm - 5am), *is_weekend*, *is_intersection*, *is_alcohol* (driver under the influence), *is_distracted*, and *is_speeding*. The class

distribution for each severity outcome was observed to be: PDO = 12,955 (64.3%), Injury = 7,083 (35.1%), and Fatal = 124 (0.6%).

METHODS

Exploratory Data Analysis & Visualization

The exploratory data analysis was done in two stages. First, temporal distributions were examined by plotting crash records across hours of day, day of week, and month. An hour-by-day-of-week heatmap was created by tabulating crash counts across all 168 hour-day combinations. Second, severity of crashes was visualized by grouping environmental conditions – weather, lighting, road surface, and type of collision – as horizontal bar charts. Crash locations were mapped spatially and a density map was generated by feeding longitudinal and latitudinal values available for independent crashes into matplotlib² as x-axis = longitude and y-axis = latitude.

ANOVA & Linear Regression

I conducted four one-way ANOVA tests to check whether mean crash severity differed significantly across levels of lighting condition, weather condition, season, and road surface condition. The outcome variable was a numeric severity score constructed by encoding PDO (Property Damage Only) = 1, Injury = 2, and Fatal = 3, reflecting the natural ordinal ordering of crash outcomes. All ANOVA computations were performed using the python library *scipy.stats* (version 1.17.1). A multiple linear regression was fitted with *persons_injured* as the outcome and the full binary predictor set mentioned earlier as regressors. The model took the standard form:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \dots + \beta_p x_p + \varepsilon$$

where $\hat{y}$ is the predicted injury count, $x_1 \dots x_p$ are the thirteen predictors, and ε is the residual error term. The data was split into a training set (80%, $n = 16,129$) and a test set (20%, $n = 4,033$). The model was fitted on the training set using ordinary least squares (OLS) via the python library *sklearn.linear_model.LinearRegression*. Standardised coefficients (β^*) were computed and the standard error was derived from the classic ordinary least squares formula,

$$\text{Var}(\beta) = \sigma^2 (X^T X)^{-1}.$$

Ordinal Logistic Regression

The centerpiece of the analysis was a proportional odds model, also known as the cumulative logit model, fit to the three-level ordinal severity outcome (PDO = 0, Injury = 1, Fatal = 2). This model was appropriate since the outcome was ordinal rather than nominal, as it respects the ordering PDO < Injury < Fatal and produces a single coefficient per predictor rather than separate coefficients per outcome level. I was unable to find standard implementations of the proportional odds model in python, and thus the model was referenced from [this paper](#) (McCullagh, P. (1980). Regression Models for Ordinal Data). The function used to compute log-likelihood was minimized using the python library *scipy.optimize.minimize*.³ I computed the standard errors and the coefficients are reported as odds ratios (OR) with 95% confidence intervals. Model fit was assessed using McFadden's R^2 and a likelihood ratio χ^2 test against the null (intercepts-only) model. The data was split 80/20 for training and testing. A full mathematical derivation of the model formulation is given in Chapter 7 of *Categorical Data Analysis, Second Ed*, by Alan Agresti.⁴

Spatial Hotspot Analysis (Supplementary)

To estimate the spatial density of crash locations across Montgomery County, a two-dimensional kernel density estimator (KDE) was applied to the GPS coordinates of all 20,162 crashes. KDE is a method that estimates the underlying probability density of a spatial point by placing a smooth kernel function over each observed point and summing the contributions across all points. A Gaussian kernel was used, which places a bivariate normal bump centered at each crash location and the KDE itself was implemented using *scipy.stats.gaussian_kde*.⁵ The critical methodological choice in KDE is bandwidth selection. A bandwidth that is too small produces a noisy, speckled surface that overfits the observed point locations; a bandwidth that is too large blurs distinct spatial structure into an uninformative plateau. Bandwidth was selected automatically using Silverman's rule of thumb⁶. Since this was outside the scope of this report, it is presented as supplementary analysis.

RESULTS

Exploratory temporal analysis revealed that Friday afternoon (3:00pm - 5:00pm) is the peak crash window ([Figure 1](#)). This is where crashes, regardless of severity, are clustered. Weekend nights (Friday to Saturday, 10pm - 2am) show a secondary elevation in crashes occurring, which is deemed to be consistent with Virginia Tech student activity patterns. When observed on a monthly window, the number of crashes peaked during September - October, which is consistent with the start of the semester and high rates of travel across traffic corridors. Crash severity was also analyzed based on environmental and road-level conditions

([Supplemental Figure 6](#)). Crashes mostly occurred in Clear/Cloudy conditions (n=16,299) and Mist/Rainy conditions produced proportionally more severe crashes as compared to Snow. Surprisingly, a Dry road surface yielded the highest proportion of crash severity, as compared to Wet, Icy, Sand/Gravel etc. On the other hand, an expected result was based on lighting conditions: Unlit and Dark roads produced higher fatality rates for crashes as compared to Daylight or Dusk/Dawn.

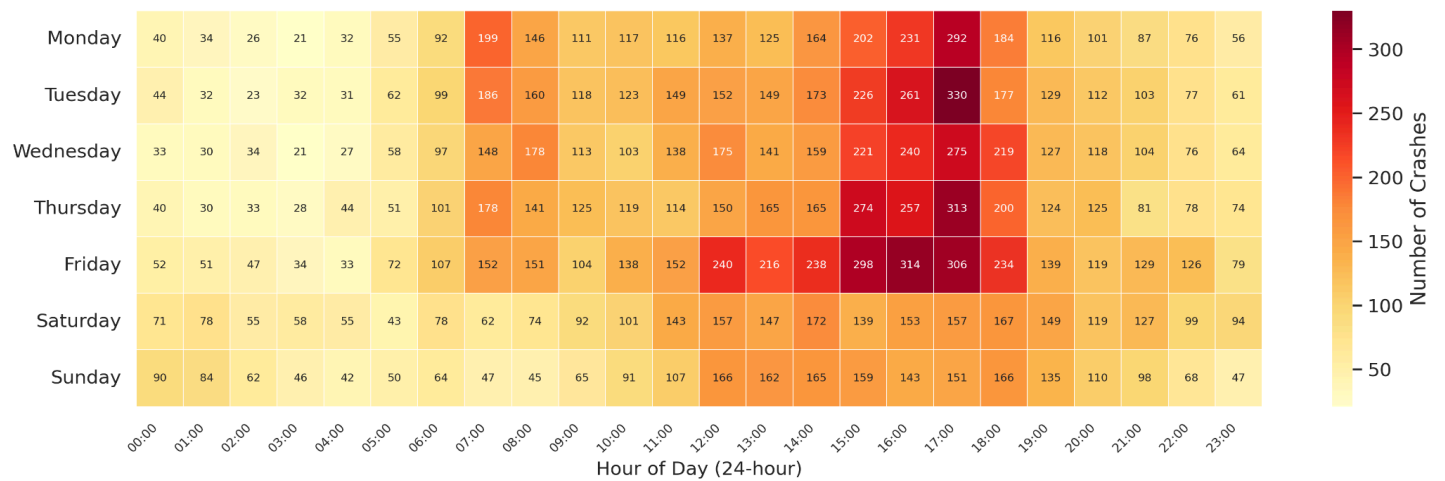


Figure 1. Crash frequency by hour of day and day of week. Cell values indicate total crash count

All four ANOVA tests returned significant at $p < 0.001$, confirming that group differences in mean severity are statistically relevant across all condition types ([Figure 2](#)). The effect sizes were uniformly small ($\eta^2 = 0.001 - 0.003$). Since I was dealing with a large sample size, $N = 20,162$, η^2 is the most informative metric, and it indicates that no single environmental or road condition dominates crash severity. Severity is instead spread out amongst all 4 types. A counterintuitive result emerged when a Tukey Honestly Significant Difference (HSD) test⁷ was used after the significant ANOVA: snowy and icy road conditions produced a lower mean severity of crash as compared to clear and dry conditions. This likely has to do with behavioral patterns of drivers who tend to reduce speed and exercise a greater deal of caution under visibly adverse conditions. Upon fitting a multiple linear regression with *persons injured* as the outcome and the full binary predictor as regressors, it was found that the model was

globally significant: $F(13, 16115) = 57.00$, $p < 0.001$. The predictors yielded the following beta coefficients: Alcohol ($\beta^* = 0.058$), intersection location ($\beta^* = 0.056$), driver distraction ($\beta^* = 0.039$), and speeding ($\beta^* = 0.042$) with each with a p-value of $p < 0.001$, making them independently significant. Night-time crashes yielded a negative beta coefficient, $\beta^* = -0.026$, $p = 0.008$, which is in-line with the behavioral driving patterns observed by the ANOVA test. A plot of the residuals vs fitted values ([Supplemental Figure 7](#)) revealed variance of residuals (errors) was unequal across observed data, resulting in a non-constant spread (“fan” shape) rather than a uniform distribution, which is expected given the zero-inflated structure of the outcome due to the extremely low percentage of fatalities. This motivates the ordinal classification, which is a better fit for an outcome with three ordered categories.

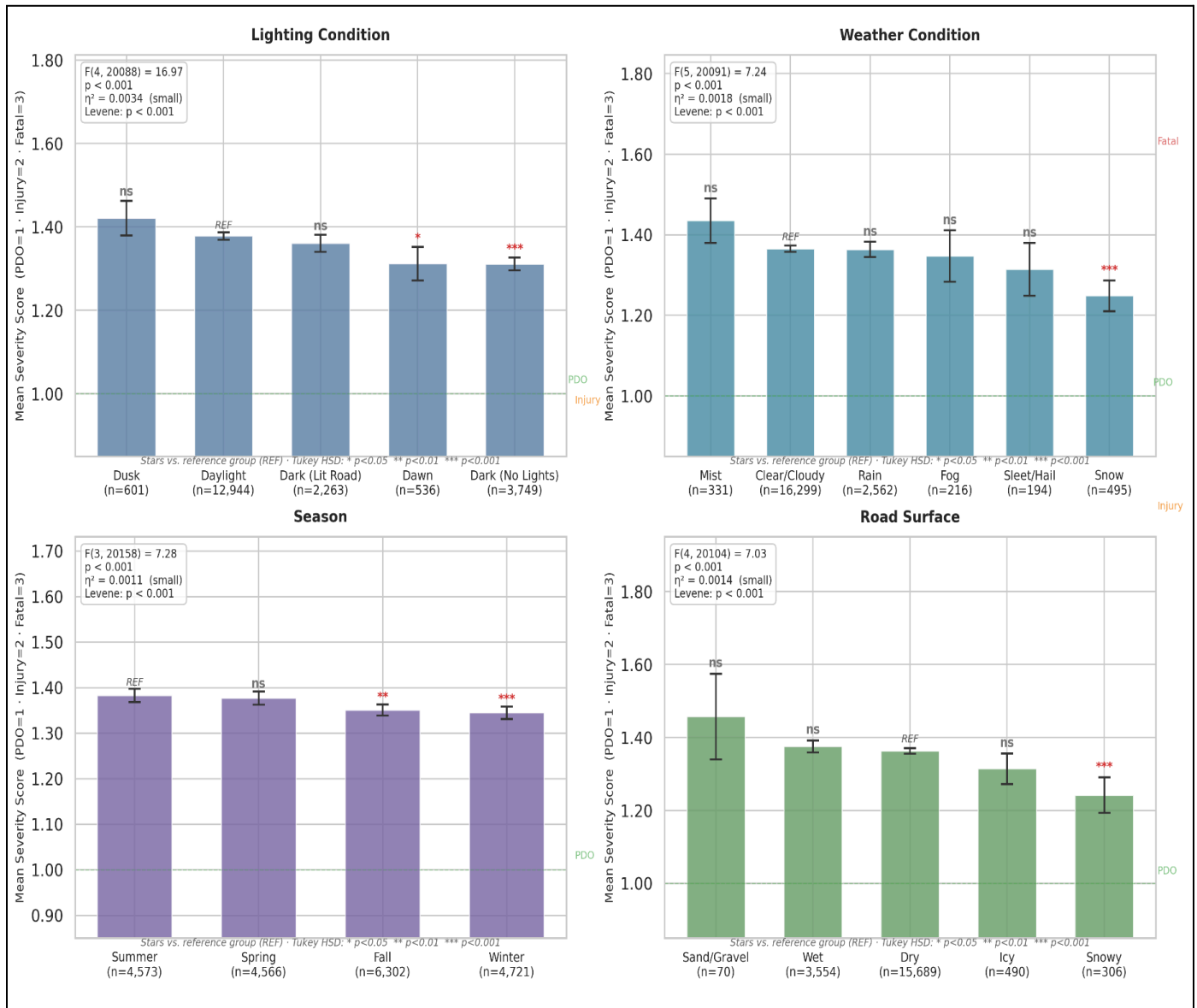


Figure 2. One-way ANOVA results: mean crash severity score by lighting, weather, season, and road surface condition. Error bars = 95% Confidence Interval. Stars indicate significance vs reference group

The ordinal logistic regression model ([Figure 3](#)) shows that Alcohol involvement was the dominant predictor (Odds-Ratio = 2.91, 95% Confidence Interval [2.45, 3.46], $p < 0.001$), with alcohol involved crashes having nearly three times the odds of belonging to a higher severity class, as compared to Adverse Weather, Night-time driving, Speeding involvement etc. Some other prominent results from the model showed that crashes at intersections were 42% more likely to be higher severity (OR = 1.42, $p < 0.001$), whereas night-time driving actually reduced severity odds (OR = 0.807, $p < 0.001$). Dark unlit roads did not yield an independent significance metric, with a p-value

of $p = 0.691$, indicating that road lighting did not have a significant effect beyond what alcohol or night-time variables already captured. On the testing data, overall classification accuracy was 64.4% and the model did not predict a fatality class, which is once again a consequence of severe class imbalance, with Fatal only contributing to 0.6% of all crashes. Upon calculating predicted probabilities ([Supplemental Figure 8](#)) for key binary predictors, it was found that alcohol and speeding produces the highest absolute shifts in Injuries and Fatalities, confirming those predictors to be the primary targets.

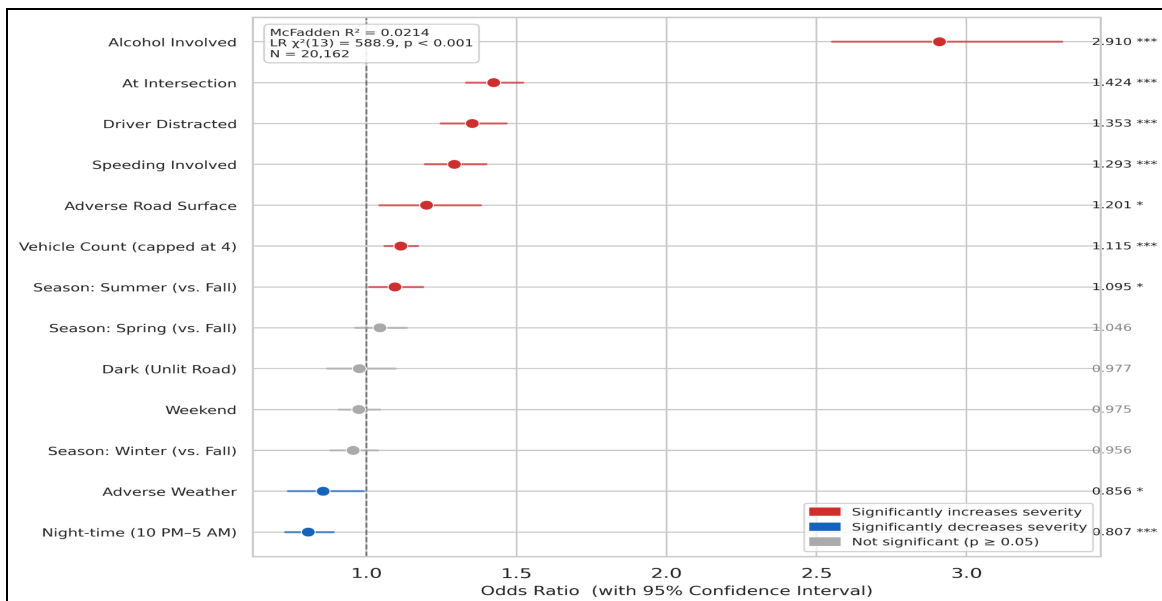


Figure 3. Proportional odds model: odds ratios with 95% confidence intervals for crash severity. Reference: Fall season, dry surface, clear weather, lit road, daytime, weekday.

SUPPLEMENTATRY SPATIAL ANALYSIS

Upon spatial analysis of where the crashes occur, I found that crashes were heavily concentrated along the US-460 highway connecting Blacksburg and Christiansburg. A secondary hotspot was revealed around Prices Fork Rd, immediately adjacent to the Virginia Tech Campus perimeter. (Figure 4) The Christiansburg commercial corridor along US-460 East forms a third distinct concentration zone, reflecting high-volume retail and through traffic. Primarily rural areas of the county show uniformly low crash density, indicating that risk and severity is not uniformly distributed across road networks, but instead channeled into a small number of high-volume traffic areas. The bandwidth for KDE was selected using Silverman's rule of thumb ($h = 0.192$).

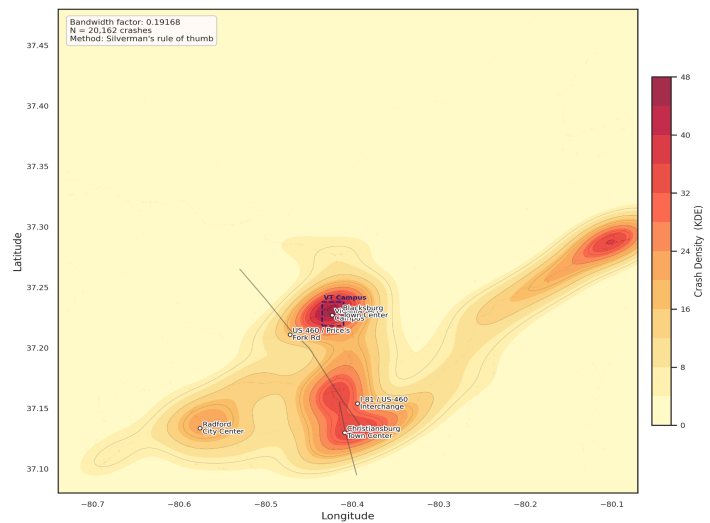


Figure 4. Kernel density estimate of crash locations, Montgomery

DISCUSSION

The central finding of this study is that crash severity in Montgomery County is determined primarily by behavioural factors – driving under the influence, speeding, and distracted driving – rather than by environmental conditions. This is consistent with national crash severity studies that I have referenced while researching the methods used in this study and is robust across the two independent models (linear regression and ordinal logistic) that are showcased.

The negative relationship between weather/night-time conditions and crash severity warrants discussion. This pattern was statistically confirmed by four one-way ANOVA tests, the linear regression and the ordinal logistic models, and lead me to conclude that it is a result of behavioral selections by drivers: deliberately reducing speed and increasing caution under visibly hazardous conditions. Similar findings have been documented in national crash datasets and studies.

Spatially, the kernel density estimation suggests that road networks in Montgomery County produce strongly channelled crash risk. The concentration along US-460 is not surprising, given its stature as the primary commercial and commuter connection between Blacksburg, Christiansburg, and Radford. As supplementary spatial context, kernel density estimation (Silverman bandwidth $h = 0.192$) reveals that crash density is not uniformly distributed across Montgomery County but is strongly channelled along the US-460 corridor and the Prices Fork Road interchange adjacent to the Virginia Tech campus.

It is important to note that the proportional odds model's $R^2 = 0.021$ reflects the randomness in individual crash outcomes. The Fatal class (which only makes up 0.6% of all records) was not reliably predicted. This is a class imbalance issue, given the large dataset of over 20,000 points. Future work leaves space for it to be addressed via oversampling methods such as Synthetic Minority Over-sampling (SMOTE)⁸.

CONCLUSIONS & LIMITATIONS

The main conclusions that I drew from the study are:

1. Alcohol involvement is the single strongest predictor of crash severity in Montgomery County ($OR = 2.91$), and this is consistent across both regression and ordinal classification.
2. Adverse weather and night-time conditions are associated with *reduced* severity – likely a behavioral effect of cautious drivers.
3. Crash locations are strongly spatially non-random, with hotspot cells concentrated along US-460, the VT campus perimeter and Radford city limits.

4. A proportional odds-ratio model implemented using ordinal logistic regression provides an interpretable, well-calibrated severity classifier for this dataset.

As it follows, limitations that affected the scope and effectiveness of this study were:

1. Fatal crashes represent only 0.6% of records ($n = 124$). This class imbalance prevents reliable Fatal prediction and limits the model's effectiveness for the highest severity outcome.
2. VT-specific factors that could have added to the spatial outlook such as game-day indicators, move-in weekend flags, etc were unavailable in the dataset and could not be included. Their omission is a known source of **residual confounding**.
3. Spatial analysis uses a regular grid created from the longitudinal and latitudinal data, rather than a road-network based unit. A network constrained KDE would be more precise and allow us to localise crash hotspots specific to different segments of the road.
4. The dataset did not include legal posted speed limits of road types (interstate vs local), which limited how fleshed out each datapoint could be.

A larger study incorporating VT-specific temporal covariates, road-network-constrained spatial analysis, and oversampling methods for the Fatal class would extend and validate these findings.

List of Supplemental Figures

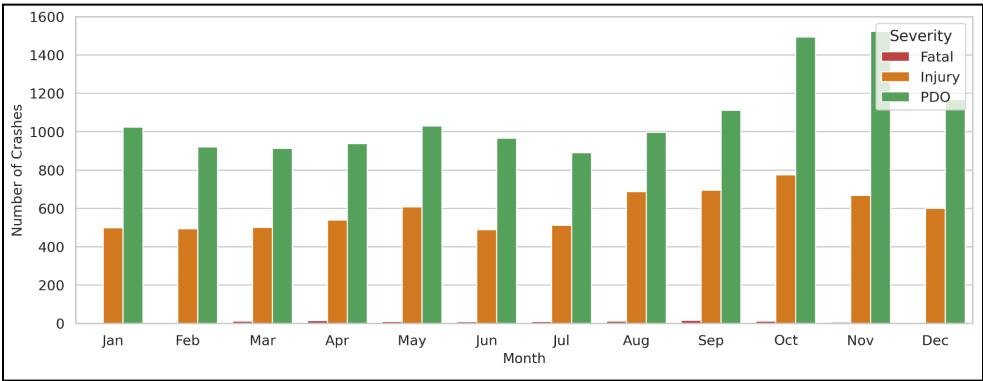


Figure 5. Crash severity proportions for each month.

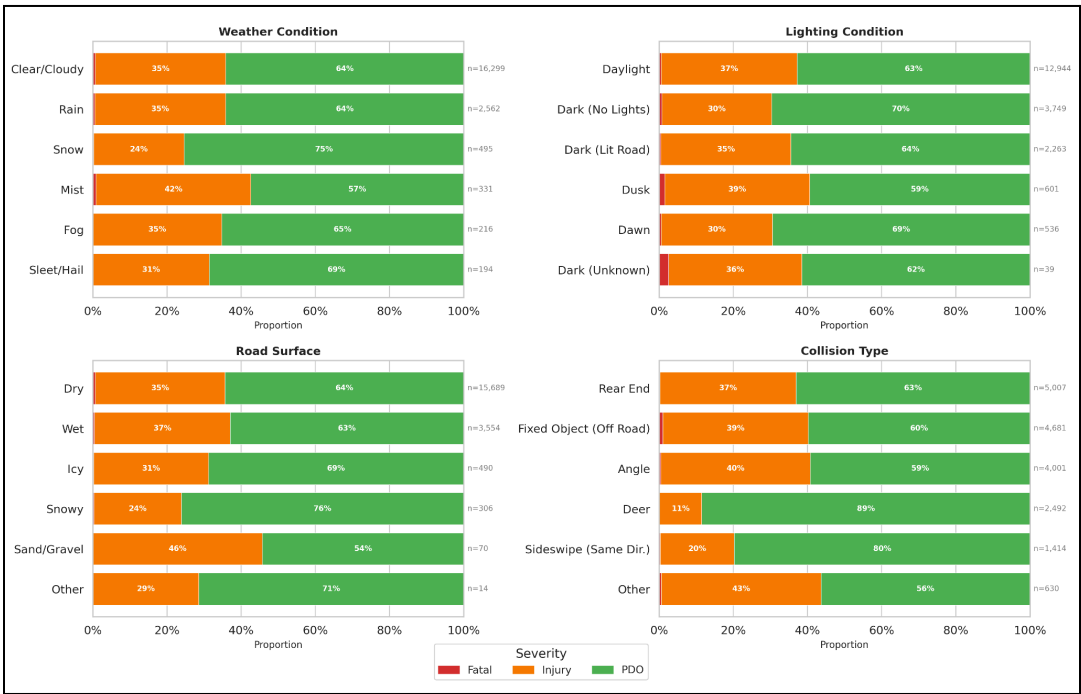


Figure 6. Crash severity proportion by environmental and road conditions. Panels show weather, lighting, road surface and collision type.

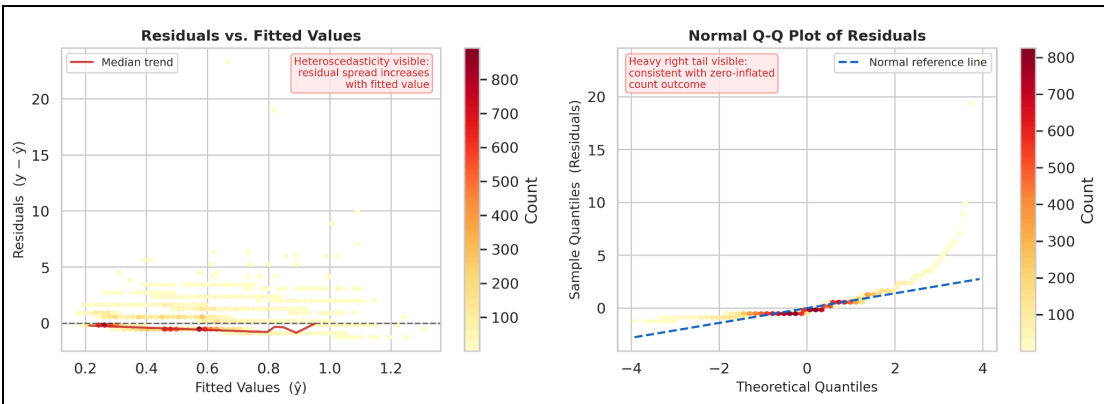


Figure 7. Regression diagnostic plots. Left: residuals vs fitted values. Right: Normal Q-Q plot showing heavy-right tail, consistent with zero-inflated counts

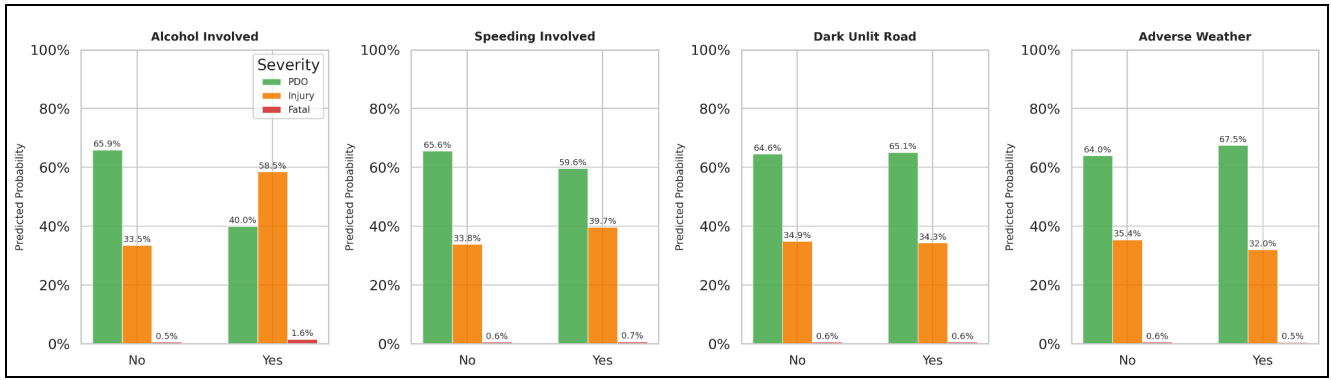


Figure 8. Predicted severity probabilities for key binary predictors, with all other predictors held at their observed mean.

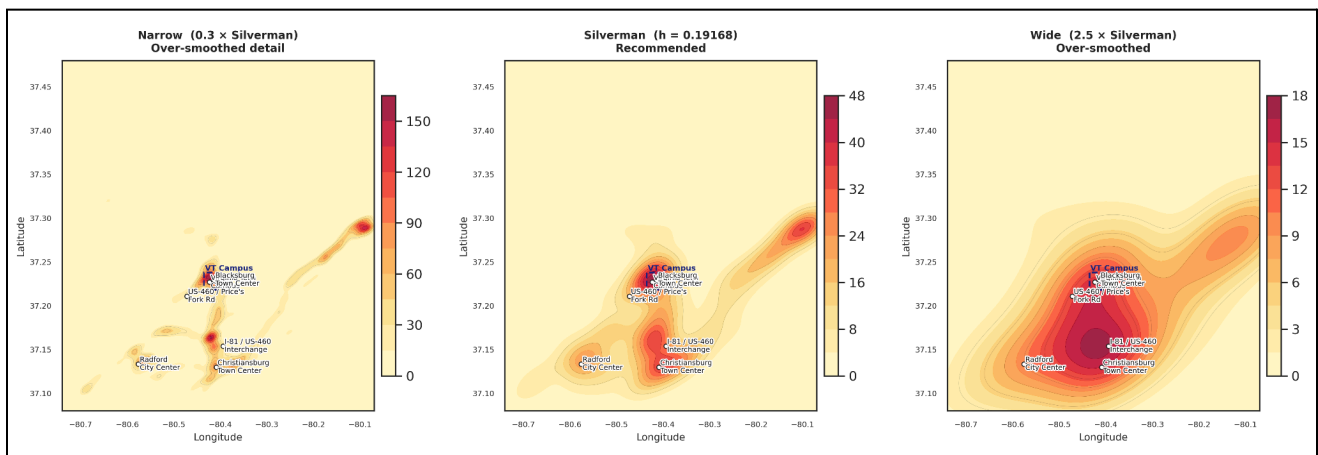


Figure 9. Three separate crash density maps showcasing how the hotspots fluctuate based on chosen Silverman's bandwidth. Left: Narrow bandwidth, Center: Default bandwidth (rule of thumb), Right: Wide bandwidth.

References

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